

Enterprise Network Design Project

Design and Implementation of a Small Enterprise Network

In Cisco Packet Tracer

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Introduction

The main goal of this project was to challenge myself by designing and implementing a network topology similar to a real-world environment, while also showcasing my skills in networking.

The idea behind this design was to create a network for a department consisting of multiple sections, each with its own requirements.

The network diagram was designed using Microsoft Visio, and the configuration and implementation were carried out in Cisco Packet Tracer.

Network Overview

This network is designed for a department consisting of several sections, including Network Manager, Public Room, Sales, IT Developers, and a Server Room.

The topology follows a hierarchical (tree-like) structure, where each section is assigned to a separate VLAN. Communication between VLANs is enabled through inter-VLAN routing.

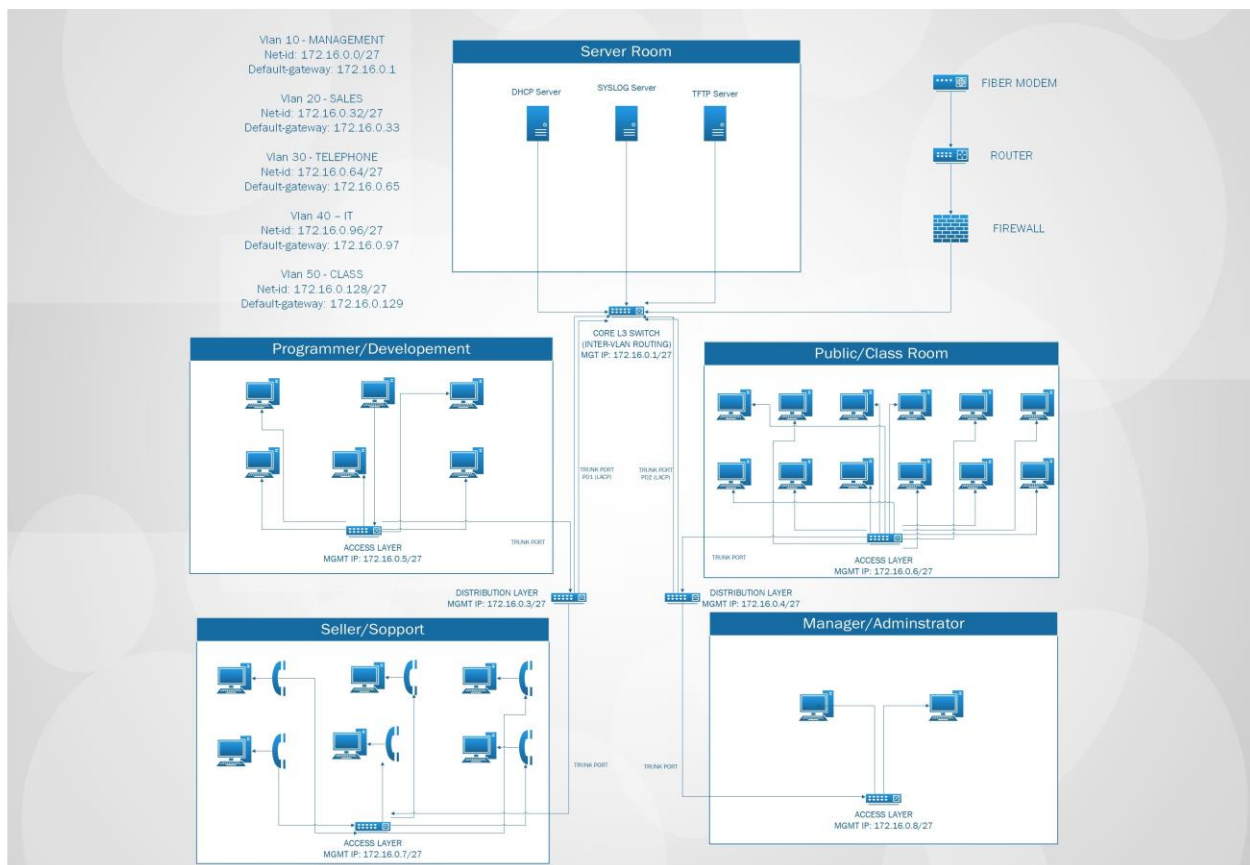
Basic security measures have been implemented within the network, and the overall design follows a three-tier architecture model, which helps improve network performance, scalability, and manageability.

Network Design

As mentioned earlier, this network is designed based on the three-tier architecture model, including Core, Distribution, and Access layers.

Although in smaller networks the Distribution layer can be omitted to reduce cost and complexity, it has been included in this design to improve scalability, simplify management, and better simulate a real-world enterprise environment.

Topology Diagram



IP Addressing & VLANs

The network is segmented into multiple VLANs, each assigned to a specific department with its own IP range and default gateway.

VLAN 10 – Management

- Network ID: 172.16.0.0/27
- Default Gateway: 172.16.0.1

This VLAN includes the Network Manager section with 2 clients that have the highest level of access. The Server Room is also part of this VLAN. Additionally, all network devices (switches) are assigned management IP addresses within this VLAN.

VLAN 20 – Sales

- Network ID: 172.16.0.32/27
- Default Gateway: 172.16.0.33

The Sales department includes 6 clients. Each user is also equipped with an IP phone.

VLAN 30 – Voice

- Network ID: 172.16.0.64/27
- Default Gateway: 172.16.0.65

Since voice traffic is sensitive to delay and requires better handling, a separate Voice VLAN is configured to support IP telephony in the Sales department.

VLAN 40 – IT

- Network ID: 172.16.0.96/27
- Default Gateway: 172.16.0.97

This VLAN is assigned to the IT department, including developers and programmers.

VLAN 50 – Public / Classroom

- Network ID: 172.16.0.128/27
- Default Gateway: 172.16.0.129

This section is designed for training sessions and can also be used as a public access area for general users within the organization.

Implementation

The implementation process started with subnetting the private IP address space. Based on the requirement of multiple sections, a /27 subnet mask was selected to properly allocate IP addresses for each VLAN.

Configuration began from the Access layer. Initial switch configurations were applied, including setting hostname, creating local users, assigning management IP addresses to VLAN interfaces, generating RSA keys, setting the domain name, and enabling SSH for secure remote access.

```
username department secret 123
enable secret 123
ip domain-name department.local
```

Most of the security measures were implemented at this layer, including:

- Disabling CDP on interfaces connected to end users
Reason: To prevent sending CDP packets to user-facing interfaces.

```
no cdp enable
```

- Port Security
Used to bind MAC addresses to switch ports and allow only authorized devices to access the network. By configuring the violation mode as restrict, the switch will log and count any unauthorized traffic on the interface.

```
switchport mode access
switchport port-security
switchport port-security mac-address sticky
switchport port-security violation restrict
```

- DHCP Snooping
Implemented to prevent DHCP spoofing attacks

```
ip dhcp snooping
ip dhcp snooping vlan [vlan-id]
trunk interface-> ip dhcp snooping trust
end user interface-> ip dhcp snooping limit rate 10
```

- Changing the native VLAN on trunk ports

To mitigate VLAN hopping attacks

```
switchport trunk native vlan 100
```

- Configuring allowed VLANs on trunk ports

To restrict traffic and allow only specific VLANs to pass through trunk links.

```
switchport trunk allowed vlan [vlan-id]
```

Sample Configurations

For instance, the following configuration represents one of the access switch interfaces in the Sales department, which connects directly to a client device:

```
switchport access vlan 20
ip dhcp snooping limit rate 10
switchport mode access
switchport voice vlan 30
switchport port-security maximum 2
switchport port-security mac-address sticky
switchport port-security violation restrict
no cdp enable
spanning-tree portfast
spanning-tree bpduguard enable
```

One of the critical configurations applied to these interfaces is enabling **portfast** and **BPDUGuard**.

PortFast is enabled to prevent issues with clients obtaining IP addresses via DHCP and to avoid generating unnecessary Topology Change Notifications (TCN), making it an essential configuration step.

BPDUGuard is a security measure. If an interface connected to a client device unexpectedly receives a BPDU message, the port will immediately enter an error-disable state.

The connection between the Access and Distribution layers is configured as trunk links, and the required VLANs are defined on each switch.

Another security measure implemented on trunk interfaces is the use of:

```
switchport nonegotiate
```

This command is used to disable DTP (Dynamic Trunking Protocol) negotiation and help prevent VLAN spoofing attacks on trunk links.

To increase bandwidth and provide better redundancy, EtherChannel was configured between the Distribution and Core layers using the LACP protocol. Two physical interfaces were bundled together as shown below:

```
channel-protocol lacp  
channel-group 1 mode active/passive
```

After configuring EtherChannel on the ports, the chosen method for load balancing is src-dst-ip. This was selected as the best option for our setup and is configured as follows:

```
port-channel load-balance src-dst-ip
```

Finally, on the Core switch, IP addresses were assigned to all VLAN interfaces (SVIs) including VLANs 10, 20, 30, 40, and 50. The IP address configured on each VLAN interface acts as the default gateway for that VLAN.

To ensure proper operation of the DHCP DORA process across multiple VLANs, the ip helper-address command was configured as shown below:

```
interface Vlan [vlan-id]
ip address [default-gateway-ip] [subnet-mask]
ip helper-address [dhcp-server-ip]
```

In addition, Syslog server addressing was configured on all switches using the following command:

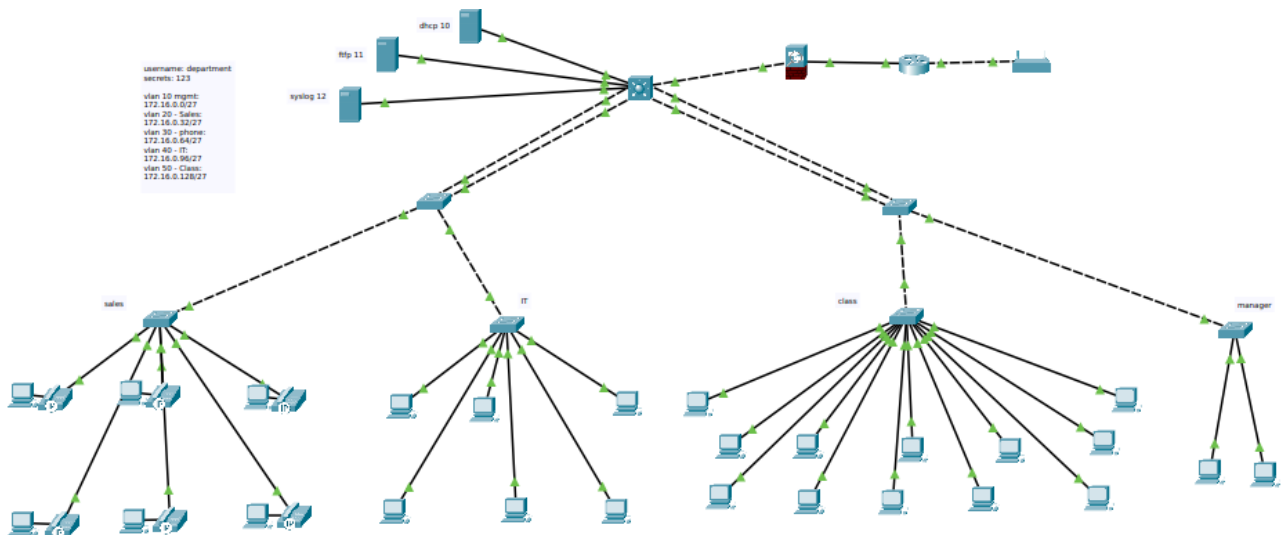
```
logging host [syslog-server-ip]
```

It is also recommended to administratively disable all unused switch ports to improve network security.

Finally, backup files from all switches were stored on the TFTP server for configuration recovery and backup purposes.

Testing and Validation

The overall network topology in Cisco Packet Tracer is represented as follows:

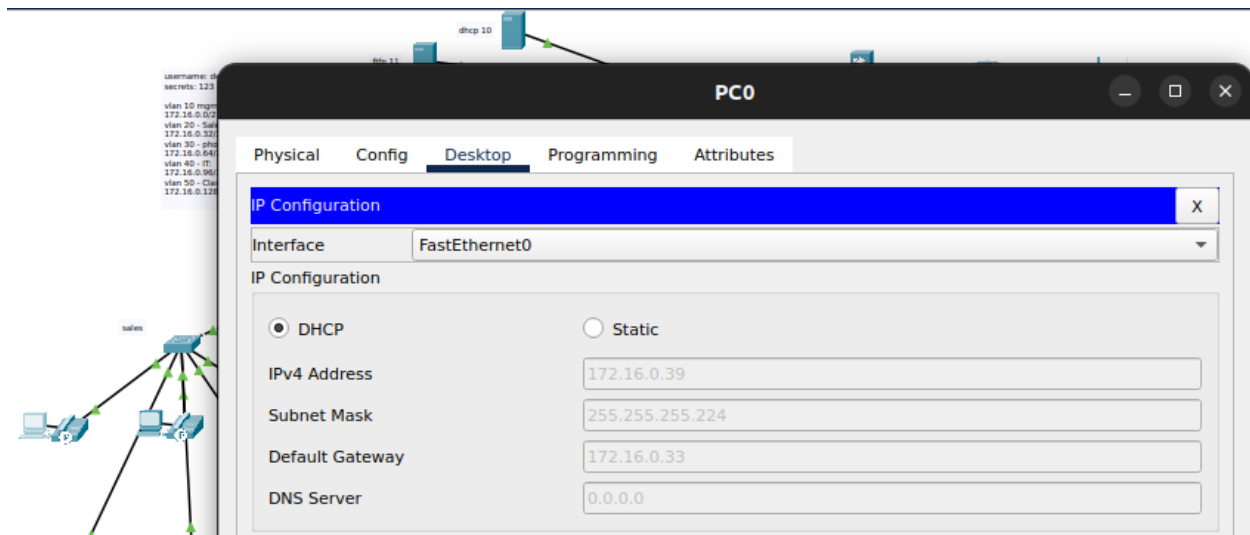
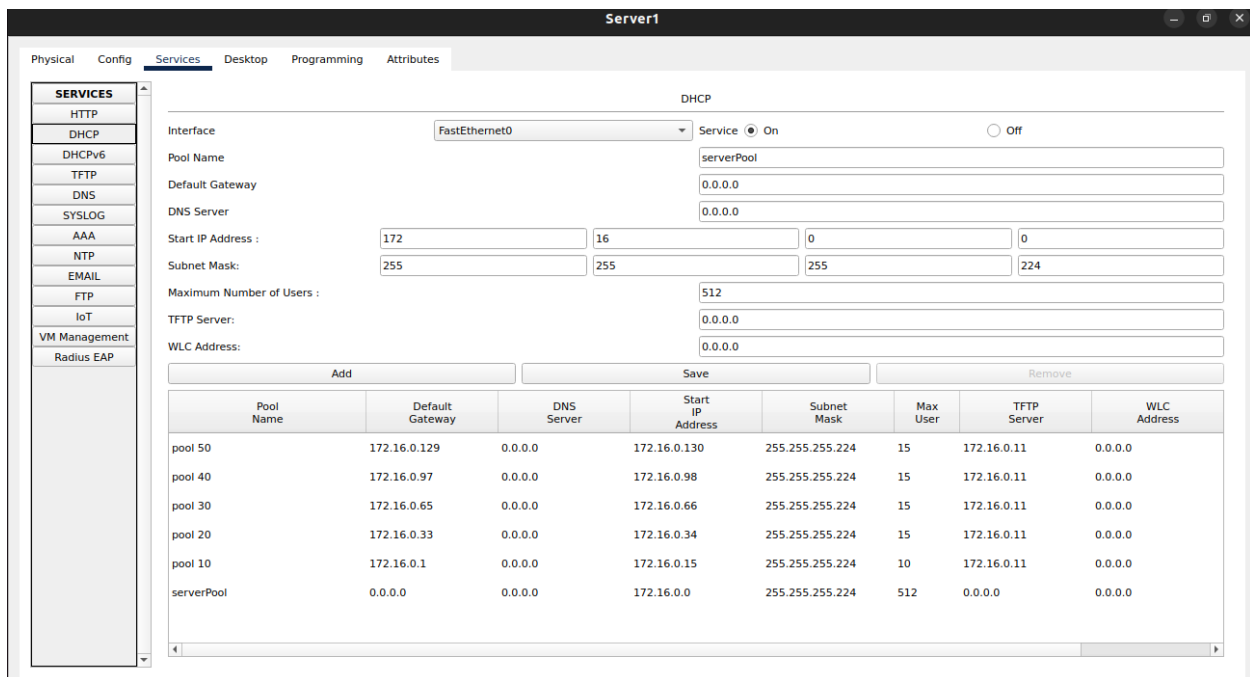


All implemented protocols and services, including DHCP, inter-VLAN communication, log management, and others, have been verified to function correctly. Below, we'll examine the successful validation of a few key aspects:

DHCP

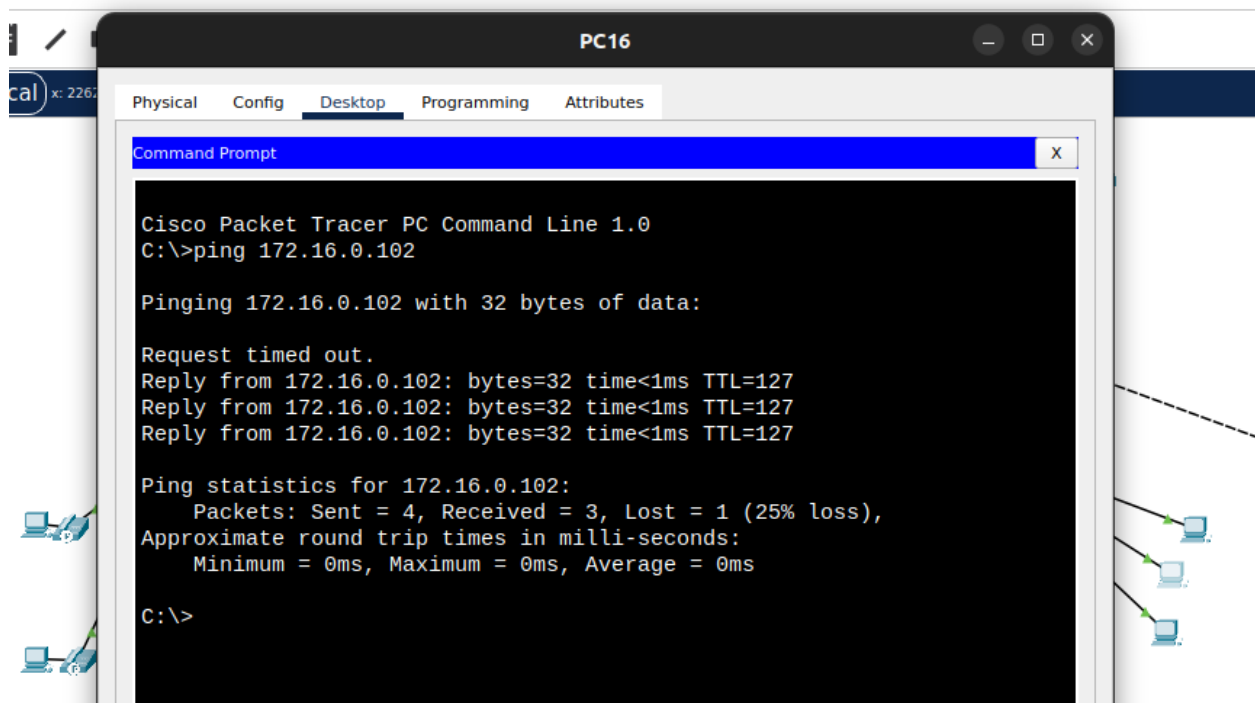
All clients successfully obtained IP addresses from the DHCP server, confirming that the DORA (Discover, Offer, Request, Acknowledge) process was completed without issues.

Additionally, the DHCP server has been configured correctly. VLAN pools have been created, and the IP address ranges have been entered.



Inter-VLAN Communication

With inter-VLAN routing implemented on the core switch, communication between VLANs is established. Successful ICMP replies will be observed when pinging any client from any other client across different segments.



Switch Log Management

Any logs generated by the switches are not only stored locally on the switch's memory but are also forwarded to the syslog server. This ensures a comprehensive and long-term repository of all switch logs is available on the server.

The screenshot displays the 'Server2' configuration window, specifically the 'Syslog' tab under the 'Services' section. The interface includes a sidebar with a list of services, a main configuration area for Syslog, and a table of log messages.

Sidebar Services:

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG**
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

Syslog Configuration:

- Service: Syslog
- Status: ☒ On ☐ Off

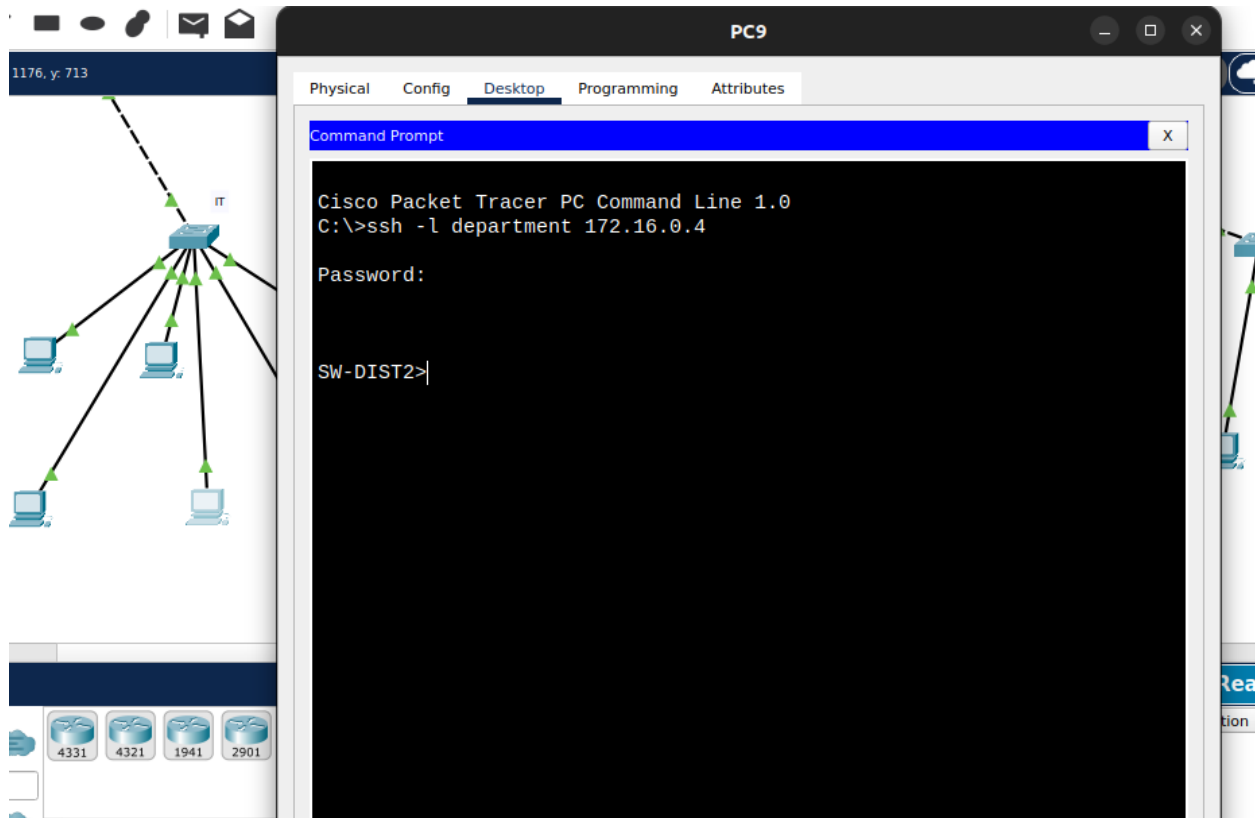
Log Messages Table:

	Time	HostName	Message
1 -		172.16.0.1	%LINEPROTO-5-UPDOWN: Line protocol on Interface ...
2 -		172.16.0.1	%LINK-5-CHANGED: Interface FastEthernet0/8, changed stat...
3 -		172.16.0.4	%LINEPROTO-5-UPDOWN: Line protocol on Interface ...
4 -		172.16.0.4	%LINK-5-CHANGED: Interface FastEthernet0/1, changed stat...
5 -		172.16.0.3	%LINEPROTO-5-UPDOWN: Line protocol on Interface ...
6 -		172.16.0.3	%LINK-5-CHANGED: Interface FastEthernet0/2, changed stat...
7 -		172.16.0.3	%LINEPROTO-5-UPDOWN: Line protocol on Interface ...
8 -		172.16.0.3	%LINK-3-UPDOWN: Interface FastEthernet0/2, changed state...

Clear Log

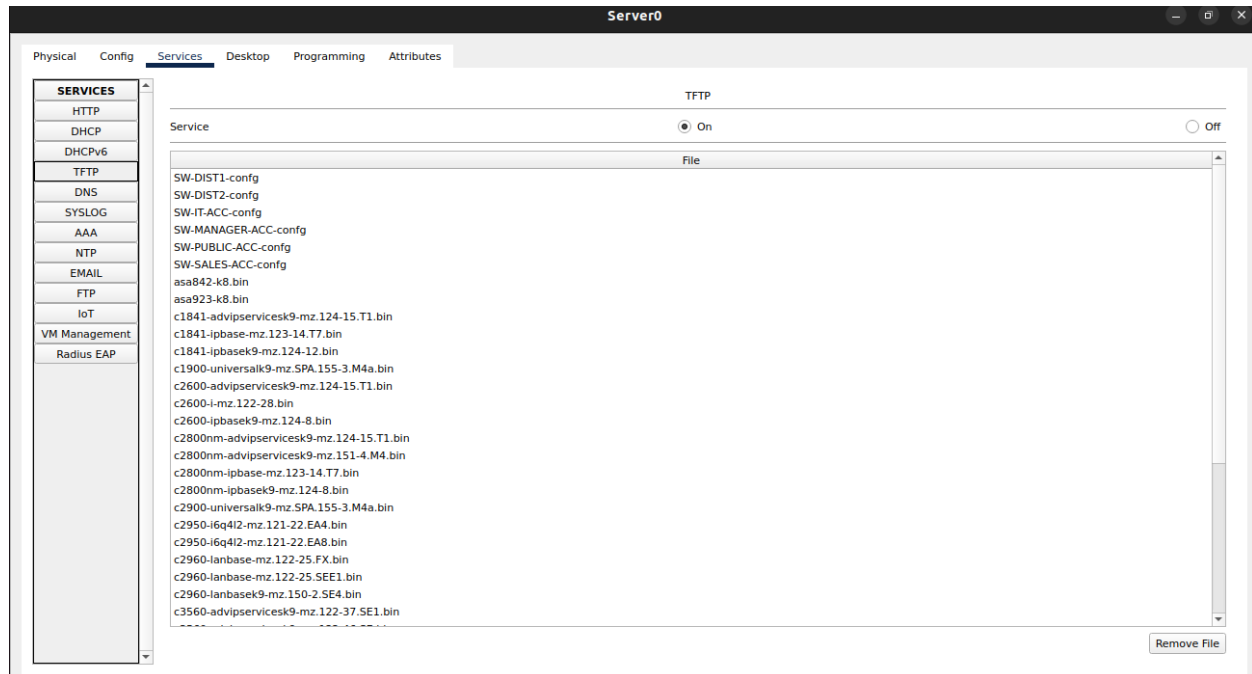
SSH Service

The SSH service is enabled on all switches, allowing for remote access via client connections. We can SSH into each switch by providing its username and IP address.



Backup Configuration

A backup configuration for each switch has been successfully taken and stored on the TFTP server. These backups are readily available for restoration in case of any issues.



Future Improvements

Although this project has been completed despite the challenges faced, there's still significant room for further development and enhancement. The areas I plan to explore and implement professionally in the near future include:

Access Control Lists (ACLs): To further bolster network security by defining specific access permissions for clients within each VLAN.

Firewall Implementation: To meticulously control incoming and outgoing traffic to the network, playing a critical role in the overall security of our local network.

Advanced Router Configuration: Enhancing my knowledge of router functionalities, including the implementation of static IP routes (IP route) and other related services on the router.

Wireless Access Point Configuration: Setting up wireless access points, similar to home Wi-Fi routers, to provide internet connectivity for wireless devices.

Conclusion

I have now reached the end of this project. This scenario served as an excellent practical exercise, allowing me to overcome its challenges and implement the design with the desired level of detail. Thank you for taking the time to read through this document.

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